

Written Assignment

Game Design

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PART-I

Q.1 The Pitch

➤ What is it?

Crash Zone: Echo Protocol is a stealth-extraction game set in a quarantined city after the crash of an unidentified extraterrestrial spacecraft. Players take on the role of a military reconnaissance operative sent into the containment zone to recover critical assets and gather intelligence about an unknown alien species.

Unlike traditional shooters where combat is the primary solution, the game's core design revolves around sound manipulation and risk management. The alien creatures have highly sensitive hearing and react aggressively to noise, making every gunshot, sprint and environmental interaction a meaningful decision.

The prototype focuses on a single mission: infiltrate the crash zone, survive hostile encounters, and extract valuable information while managing the consequences of the noise you create.

➤ Who is it for?

The game is aimed at players who enjoy tactical decision-making, stealth gameplay and high-tension encounters. It is designed for fans of extraction and survival experiences who prefer outsmarting threats rather than simply overpowering them.

The target audience includes players who enjoy:

- Tactical shooters and extraction games
- Stealth-focused gameplay
- Risk-versus-reward decision making
- Exploration and mystery-driven progression
- Emergent gameplay systems

➤ Why would someone play it?

Most stealth games revolve around avoiding an enemy's vision. In Crash Zone: Echo Protocol, the primary resource is sound.

Players are constantly making meaningful decisions:

- Is firing my weapon worth attracting nearby enemies?
- Should I sprint to safety or stay hidden?
- Can I complete the mission without revealing my position?
- What can I learn about the alien threat before extracting?

The appeal comes from mastering the relationship between player actions and enemy behaviour. Every encounter becomes a small tactical puzzle where knowledge is often more valuable than firepower.

Beyond the immediate gameplay, the game's mystery-driven premise encourages players to uncover more information about the crash, the alien species, and humanity's response to the event. The combination of tension, discovery and systemic decision-making creates a gameplay experience that rewards curiosity and intelligent play rather than raw combat skill.

Q.2 Core Loop & First Session

Core Gameplay Loop: The core loop is built around observation, decision-making and extraction.

Deploy into Crash Zone



Observe enemy patrols and identify safe routes



Navigate through hostile territory using movement and cover



Manage noise through walking, sprinting, crouching and combat decisions



Reach mission objective



Recover intelligence or critical assets



Extract safely



Unlock new information about the alien threat

The loop is designed around the idea that information is more valuable than firepower. Every action creates risk, and players are rewarded for understanding enemy behaviour rather than simply eliminating threats.

First Session Experience: The first session is designed to teach players the game's central rule:

"Sound has consequences."

Players begin in a small section of the quarantine zone with a simple objective:

"Recover a classified asset and extract."

During the first few minutes, players naturally discover:

- Movement options such as walking, sprinting and crouching
- The effectiveness of using cover
- The presence of hostile alien patrols
- The relationship between sound and enemy behaviour

Most players will instinctively fire their weapon when they encounter danger. This creates the game's first learning moment. The gun solves the immediate problem, but the resulting noise attracts additional enemies, teaching players that combat is a **tactical trade-off**, rather than a universal solution. As players progress through the level, they begin to understand that avoiding attention is often safer than engaging every threat.

➤ **What Brings Them Back Tomorrow?**

The Day-1 hook is 'Curiosity'.

By the end of the mission, players successfully recover an asset from the crash zone, but instead of answering questions, it raises new ones. The recovered intelligence hints that the alien presence is more organized than initially believed and that the crash itself may not have been accidental.

Players leave their first session wondering:

- What exactly are these creatures?
- Why did they arrive?
- What happened at the crash site?
- What other secrets exist within the quarantine zone?

Alongside the narrative mystery, players are motivated by mastery. The first session teaches the basics of survival, but players quickly realize there are more efficient and intelligent ways to complete missions. This creates a strong desire to replay encounters, improve strategies and better understand the enemy ecosystem.

The combination of mystery, player mastery and Risk Vs. Reward decision-making forms the foundation for long-term engagement.

Q.3 Progression & Metagame

➤ Day-1 Hook

The primary Day-1 hook is the mystery surrounding the alien crash and the realization that the player understands very little about the threat they are facing.

The first mission ends with the recovery of a classified asset from the crash zone. Instead of providing clear answers, the recovered data reveals evidence of structured alien behaviour, encrypted signals and indications that the crash may not have been accidental.

This creates an immediate question in the player's mind:

"What is actually happening inside the quarantine zone?"

The goal of the first session is not to answer this question, but to create enough curiosity that players want to return for another mission.

➤ Progression System

Progression is built around two parallel systems:

1. Knowledge Progression

As players complete missions, they recover intelligence, recordings, biological samples and transmission fragments. Each discovery expands humanity's understanding of the alien species and unlocks new information about:

- Enemy behaviour patterns
- Detection methods
- Communication systems
- Environmental hazards
- Crash site mysteries

Knowledge becomes a form of progression. Players are rewarded not only for surviving missions but also for learning how the world works.

2. Equipment Progression

Recovered intelligence allows military researchers to develop new tools for future operations.

Examples include:

- Improved sound dampening equipment
- Enhanced reconnaissance devices
- Better detection tools
- Advanced distraction gadgets
- Tactical movement upgrades

These upgrades expand player options rather than simply increasing damage numbers.

➤ Long-Term Metagame

The long-term metagame revolves around uncovering the truth behind the crash and improving humanity's ability to operate inside the quarantine zone.

Each mission contributes to a larger investigation. Players gradually piece together the origins, intentions and behaviour of the alien species while unlocking new regions, mission types and strategic tools.

The objective is to create a progression system where players feel that every successful extraction contributes to a larger goal.

➤ **What Keeps Players Playing For Months?**

Three factors drive long-term engagement:

Mystery: Players want to uncover the full story behind the crash, the alien species and the events occurring within the quarantine zone.

Mastery: Players continuously improve their understanding of enemy behaviour, noise management, route planning and mission execution. The better they understand the systems, the more efficiently they can complete objectives.

Strategic Growth: As new tools and information become available, players gain access to new approaches and strategies. Rather than simply becoming stronger, they become more capable and informed operators.

Together, these systems create a progression structure where curiosity drives short-term retention, mastery drives medium-term engagement and long-term investigation provides a reason to continue playing for months.

Q.4 Monetization

My monetization philosophy would be to sell self-expression rather than player power.

The core gameplay of **‘Crash Zone: Echo Protocol’** is built around information gathering, stealth and decision-making. Players should succeed because they understand the game's systems, not because they spent more money than other players. For that reason, I would avoid selling power, combat advantages or progression shortcuts that undermine the core experience.

➤ **Cosmetic Customization:**

The primary monetization pillar would be cosmetic content, including:

- Recon operator skins
- Weapon skins and visual variants
- Character animations and emotes
- Extraction effects
- Profile customization options

These purchases allow players to personalize their experience without affecting gameplay balance.

➤ **Seasonal Operations Pass**

A seasonal progression pass would provide a structured way for players to earn cosmetic rewards while participating in ongoing investigations within the quarantine zone.

Each season could introduce:

- New narrative discoveries
- Additional lore fragments
- Limited-time cosmetic rewards
- Seasonal challenges

This keeps players engaged while providing recurring monetization opportunities.

➤ **Premium Mission Packs**

Rather than selling power, the game could offer optional premium mission campaigns that expand the story and provide additional investigation opportunities.

These missions would deliver:

- New environments
- Unique objectives
- Additional lore
- Cosmetic rewards

Players who enjoy the mystery and world-building could choose to purchase more content without creating an advantage over others.

➤ **Why It Doesn't Feel Like a Cash Grab?**

The most important principle is that money should never replace skill.

Players who spend money should gain access to personalization, convenience, and additional content, but they should not gain a competitive advantage. The game's core appeal comes from understanding enemy behaviour, managing risk, and making intelligent decisions. Preserving that relationship is critical to maintaining player trust.

My goal would be for players to feel that purchases enhance their experience rather than solve problems intentionally created by monetization.

Q.5 Use of AI

AI played a significant role throughout the development process, both as a prototyping accelerator and as a design collaborator.

➤ **How I Used AI During Development?**

I primarily used AI as a rapid iteration tool rather than a content generation tool. During the concept phase, AI helped me explore multiple genres, challenge weak ideas, evaluate scope and identify a prototype direction that could realistically be built within the assignment timeline. Instead of immediately committing to the first idea, I used AI to compare different approaches, identify risks and refine the core player experience.

Once development began, AI assisted with:

- Gameplay system brainstorming
- Prototype scoping
- Core loop validation
- Technical problem-solving in Unity
- Code debugging and implementation support
- Documentation and design review

Rather than replacing design decisions, AI allowed me to iterate on those decisions faster. Every major gameplay choice, including the stealth-extraction structure, sound-based threat system and progression concepts was evaluated and refined through multiple rounds of discussion and testing.

In many ways, I treated AI as a design partner that could continuously challenge assumptions, propose alternatives and help identify blind spots throughout development.

➤ **Is AI Part of the Game Itself?**

For the prototype, AI is not directly implemented as a player-facing feature. My focus was validating the core gameplay loop before introducing additional complexity. However, if developed further, AI would become an important part of the game's long-term design.

One direction I find particularly interesting is adaptive enemy behaviour. Instead of relying on completely scripted encounters, alien entities could gradually learn from player actions and adjust their patrol routes, investigation priorities and threat responses over time.

For example:

- Frequently used routes may become more heavily monitored.
- Common distraction locations may become less effective.
- Enemy patrol patterns could evolve based on player behaviour across multiple missions.

This would reinforce the core fantasy of facing an intelligent and poorly understood species while ensuring that players continue adapting their strategies instead of relying on repetitive solutions.

➤ Why AI Matters To This Project?

The most valuable aspect of AI was not automation but iteration speed.

As a solo developer working within a limited timeframe, AI allowed me to spend less time searching for solutions and more time testing ideas. This made it possible to focus on the most important question behind the assignment:

- I believe AI is most powerful when it amplifies human creativity and decision-making rather than replacing it. Throughout this project, it served as a tool that accelerated experimentation, improved productivity and helped refine the final design.

Q. 6 Shipping

➤ What Would I Test First?

The first thing I would test is whether the sound-driven stealth system creates meaningful player decisions. The entire game is built around a simple hypothesis:

“Players will enjoy manipulating enemy behaviour through sound more than simply eliminating enemies through combat.”

Before investing further into content, progression systems, narrative or monetization, I would run playtests focused on this mechanic alone.

Key questions I would try to answer:

- Do players understand that sound attracts enemies?
- Do players intentionally change their behaviour because of it?
- Do players view combat as a tactical choice rather than the default solution?
- Does successfully avoiding detection feel rewarding?

If the sound system fails to create interesting decisions, the rest of the game loses its foundation.

➤ **What Would Make Me Kill The Project?**

The strongest reason to stop development would be if players consistently ignored the stealth systems and found direct combat more enjoyable.

For example:

- Players intentionally create noise because there is little downside.
- Players solve every encounter by shooting.
- Players do not understand how enemy detection works.
- The stealth mechanics feel frustrating rather than strategic.

In that situation, I would either redesign the core mechanic or pivot the project toward a more action-oriented experience. I would rather discover that problem early than continue building content around a weak foundation.

➤ **What Would Make Me Change Direction?**

A pivot would be considered if testing revealed that players enjoyed the information-gathering and investigation aspects more than the stealth itself.

In that scenario, I would consider increasing the focus on reconnaissance, discovery and adaptive enemy behaviour while reducing emphasis on combat encounters.

The goal would be to follow what players genuinely enjoy rather than forcing them into a predetermined playstyle.

➤ **Metrics I Would Watch During Soft Launch**

Rather than tracking a large number of metrics, I would focus on a small set that directly measures whether the core design is working.

1. Mission Completion Rate: This tells me whether players can successfully understand and complete the mission.

A very low completion rate may indicate confusion, difficulty spikes or poor onboarding.

2. Stealth Usage Rate: I would track how often players crouch, avoid enemies, or bypass encounters compared to direct combat. This metric helps determine whether players are engaging with the intended gameplay loop.

3. Repeat Mission Rate: I would measure how many players start another mission after completing their first one. This is the clearest indicator of whether the core loop is compelling enough to generate immediate retention.

➤ **Success Criteria**

The project would be considered successful if players:

- Understand the sound system quickly.
- Modify their behaviour because of it.
- Feel rewarded for avoiding unnecessary combat.
- Choose to play another mission after their first successful extraction.

If those outcomes are achieved, I would have confidence that the core loop is strong enough to justify further development.

Q.7 Reference Games & Inspirations

This project was inspired by a combination of film and game references, with each contributing to a different part of the design.

➤ **Film Inspirations:**

World War Z

World War Z heavily influenced the enemy behaviour design. While the prototype does not use zombies, I was inspired by the idea of creatures reacting aggressively to sound and the tension that emerges when every noisy action can attract danger. The concept of treating sound as a threat multiplier became one of the foundations of the gameplay loop.

Battleship

Battleship inspired both the setting and the perception model of the enemy species. I was particularly interested in the idea of humanity responding to an unfamiliar extraterrestrial threat while trying to understand its capabilities. The film's concept of enemies operating with different detection methods influenced the decision to build creatures that rely primarily on hearing rather than traditional vision.

➤ Game Inspirations

Alien Isolation

Alien Isolation influenced the overall feeling of vulnerability and tension. Rather than creating an enemy that exists only to be defeated, I wanted the player to feel that avoiding detection was often the smarter choice. The game's emphasis on observation and survival strongly influenced the intended player experience.

Escape from Tarkov

Escape from Tarkov inspired the extraction structure of the game. The idea of entering a dangerous area, accomplishing an objective, and successfully leaving with valuable information became the foundation of the mission loop. I was particularly interested in the risk-versus-reward decisions that extraction-based gameplay naturally creates.

Metal Gear Solid V

Metal Gear Solid V influenced the stealth design philosophy. Its emphasis on player creativity, multiple approaches to objectives, and emergent solutions helped shape my approach to mission design. Rather than prescribing a single correct path, I wanted players to evaluate situations and decide how to navigate them using the tools available.

➤ What I Did Differently:

Most stealth games focus on avoiding an enemy's vision, while most shooters reward direct confrontation. My goal was to combine stealth and extraction gameplay around a different core resource: **'Sound'**.

In this design, every significant action creates a consequence. Sprinting, shooting and movement all influence enemy behaviour, turning sound into a strategic resource that players must actively manage. The result is a stealth-extraction experience where understanding and manipulating enemy behaviour is often more valuable than eliminating threats through combat.

PART-II

Q. Gameplay Insights & Strategy

Genre: Information-Driven Survival Games

This genre I believe remains largely unsolved. These are games where survival, exploration and decision-making are central to the experience, but where player knowledge has not yet become the primary progression system.

Examples include:

- Project Zomboid
- DayZ
- State of Decay 2

All three games create memorable survival stories and strong emergent gameplay. However, their long-term progression eventually shifts toward familiar goals such as acquiring better equipment, securing resources, expanding a base or increasing combat effectiveness. The player's understanding of the world improves naturally, but the game rarely treats knowledge itself as the main reward.

The biggest challenge is that information is difficult to package as progression.

Traditional free-to-play systems are built around visible rewards:

- Higher levels
- Better weapons
- Stronger characters
- Larger numbers

Knowledge is much harder to communicate and monetize because it exists inside the player's head rather than inside the game's economy. As a result, most games eventually fall back to gear progression even when their most interesting moments come from discovery and learning.

➤ **What A Breakout Hit Would Need:**

I believe a breakout title would need to make information a first-class progression system.

Instead of asking: **"How much stronger is my character?"**

Players should ask: **"What do I know now that I didn't know before?"**

Examples include:

- Discovering enemy behaviour patterns
- Learning safe routes through dangerous areas
- Uncovering hidden relationships between game systems
- Building a shared intelligence network with other players
- Unlocking strategic opportunities through observation rather than combat

The reward loop would focus on understanding rather than accumulation.

Mobile free-to-play is uniquely positioned to support this idea because of its session structure.

Short play sessions naturally encourage:

- Exploration
- Investigation
- Discovery
- Experimentation

Rather than long grinding sessions, The challenge is building progression systems that make information feel as rewarding as traditional upgrades.

➤ **My Vision:**

The breakthrough version of this genre would combine:

- Survival tension
- Persistent world discovery
- Social knowledge sharing
- Dynamic enemy behavior

Players would not simply become stronger over time. They would become smarter. That distinction sounds small, but I believe it represents one of the largest untapped opportunities in free-to-play game design today.

PART-III

Q. Design Specification

➤ Feature: Tactical Forecast System

One of the strengths of survivor-like games is the constant stream of meaningful upgrade choices. However, in many existing titles, players make these choices with limited information about what challenges lie ahead.

As a result, upgrade decisions often become reactive rather than strategic.

I chose the Tactical Forecast System because it introduces planning and anticipation into the genre without changing the core accessibility that makes survivor-like successful. The feature rewards player decision-making rather than simply increasing complexity.

➤ Feature Overview:

At specific intervals during a run, players receive intelligence reports about upcoming enemy waves. These reports provide partial information about future threats, allowing players to make more informed upgrade decisions.

Examples:

- Heavy enemy concentration detected.
- Elite enemy approaching.
- Increased ranged threats expected.
- High-speed swarm incoming.
- Boss mutation likely.

The information is intentionally incomplete. Players receive enough insight to plan, but not enough to remove uncertainty.

➤ Design Intent:

The goal is to transform upgrade selection from a purely reactive system into a strategic one.

Instead of asking: **"What is my strongest upgrade?"**

Players begin asking: **"What do I need for the next challenge?"**

This creates more meaningful decision-making while preserving the fast-paced nature of survivor-like gameplay. The system also increases build diversity because players may choose situational upgrades that directly counter expected threats rather than always selecting the highest damage option.

➤ **Player Decision Flow:**

The most important design objective is creating a stronger relationship between information and player choice.

Consider the following scenario:

The player reaches Level 12 and receives a forecast warning:

"Large armored enemy group detected."

Immediately afterward, the player is offered three upgrades:

1. Increased Area Damage
2. Increased Attack Speed
3. Increased Armor Penetration

Without the forecast system, many players would default to familiar choices such as attack speed.

With the forecast system, armor penetration becomes significantly more attractive because the player understands the challenge ahead.

This creates an important psychological shift.

The player no longer feels that success came from luck or from selecting the strongest upgrade. Instead, success comes from interpreting information and making an informed decision.

If the player survives the upcoming wave, they experience a sense of validation:

"I predicted the threat correctly."

That feeling is far more satisfying than simply watching larger damage numbers appear on screen.

Over multiple runs, players begin learning enemy behaviours, wave patterns, and build synergies. This creates a deeper mastery layer without increasing mechanical complexity.

Wireframe 1: Incoming Threat Warning

INCOMING		THREAT	REPORT
Armored	Units	Detected	
Estimated	Arrival:	45	Seconds
Threat	Level:		High
[CONTINUE]			

Wireframe 2: Upgrade Selection

CHOOSE	AN	UPGRADE
[Area	Damage	+20%]
[Attack	Speed	+15%]
[Armor	Penetration	+25%]
Recommended		Against:
Armored		Units

Wireframe 3: Outcome Feedback

WAVE		COMPLETE
Forecast	Counter	Successful
Armored	Threat	Neutralized
Bonus	XP	Earned

➤ **Expected Impact:**

The Tactical Forecast System adds strategic depth without disrupting the simplicity that makes survivor-like games accessible. By giving players information before major decisions, the system encourages planning, experimentation and mastery while preserving the excitement of adapting to unpredictable situations.

The result is a survivor-like experience where player knowledge becomes as important as player progression.